

Activity 1

Math 2480

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1 Part 1: Summarizing One Categorical Variable

1.a Frequency Table

Color	Frequency	Relative Frequency
BLACK	44	0.13622291
BLUE	37	0.11455108
GOLD	40	0.12383901
GRAY	47	0.14551084
GREEN	21	0.06501548
RED	15	0.04643963
SILVER	74	0.22910217
WHITE	45	0.13931889

1.b Appropriate Plot for Summary of Color

An appropriate plot for summarizing a categorical variable like color is a bar chart. Each bar represents a different color, and the height of the bar corresponds to the frequency or relative frequency of that color in the dataset.

1.c Bar Chart of Color Counts

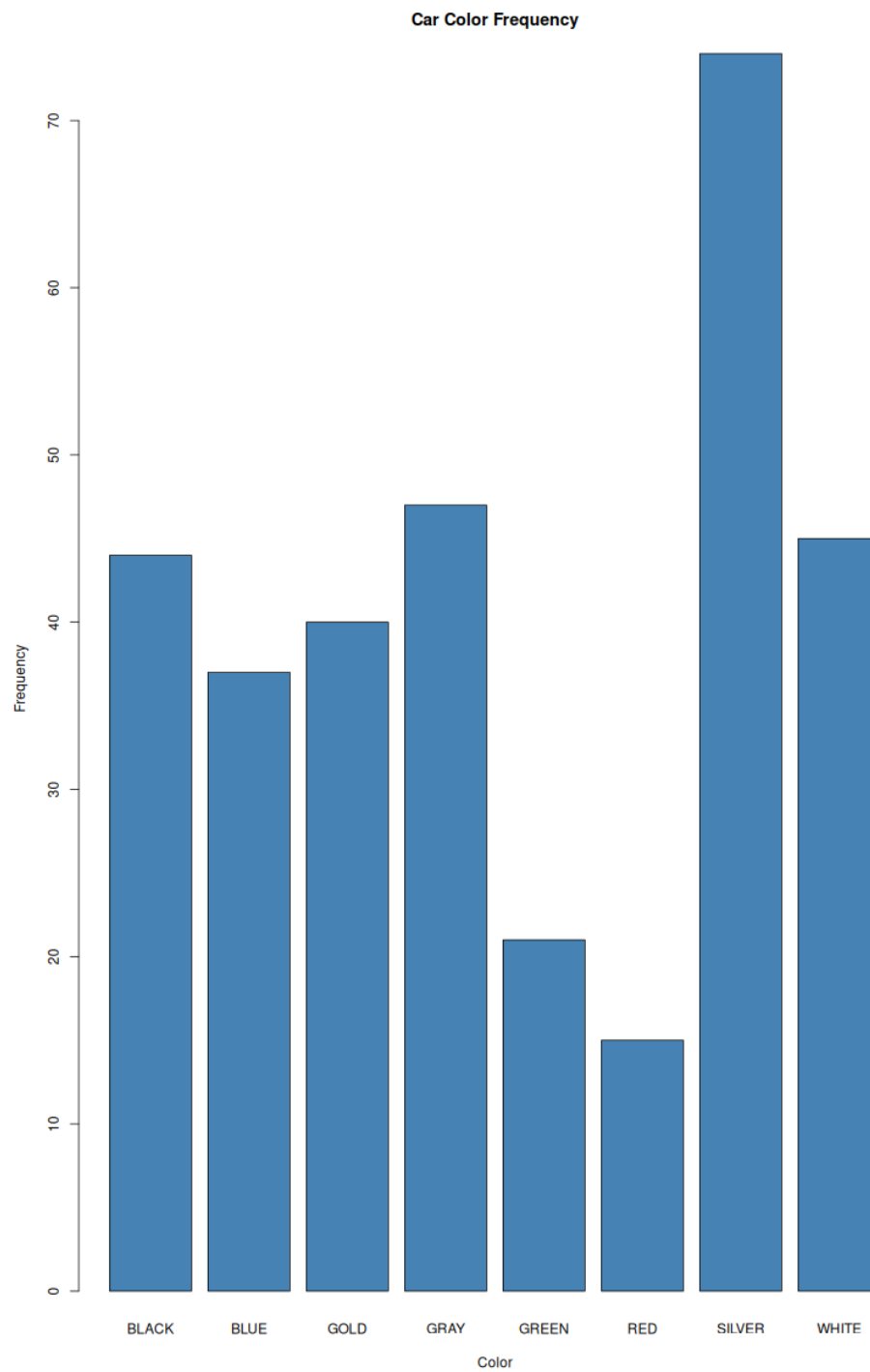


Figure 1: Bar Chart of Color Counts

2 Summarizing One Quantitative Variable

2.a Mean and Variance

Mean	11754.1
Variance	7648459

2.b Five Number Summary

Minimum	5995
First Quartile (Q1)	9967.5
Median (Q2)	11000
Third Quartile (Q3)	14500
Maximum	20995

2.c Boxplot of Price

The outlier in this dataset is located at \$20,995.



Figure 2: Boxplot of Price

2.e Log Transformation of Price Boxplot

New outlier of 8.698681.

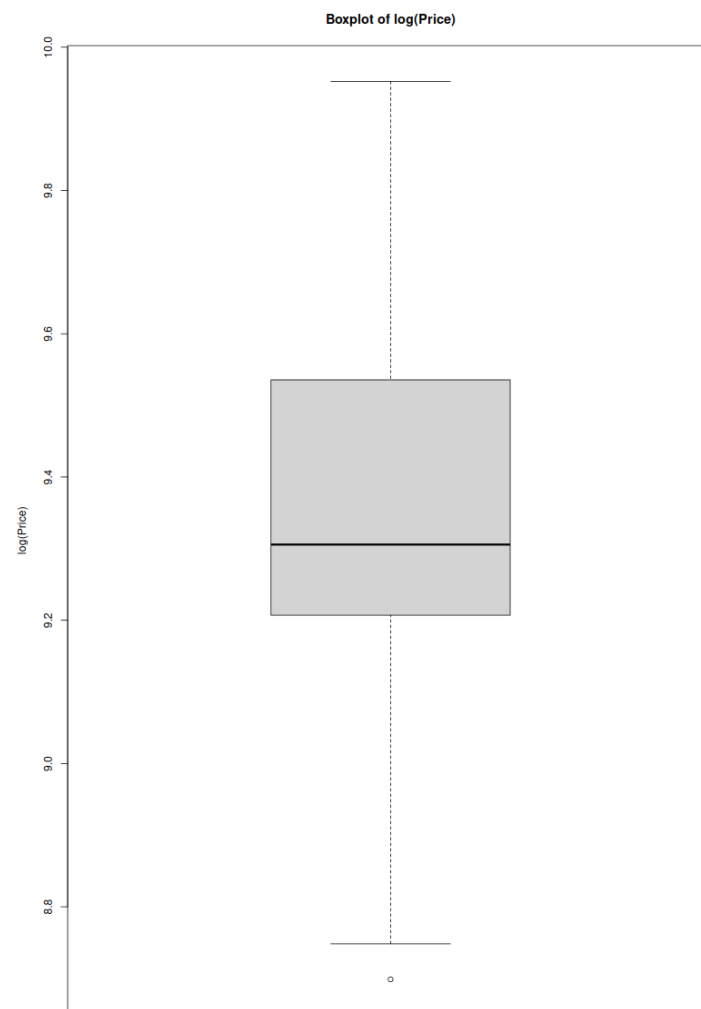


Figure 3: Boxplot of Log-Transformed Price

2.f Normal and Log Price Histograms

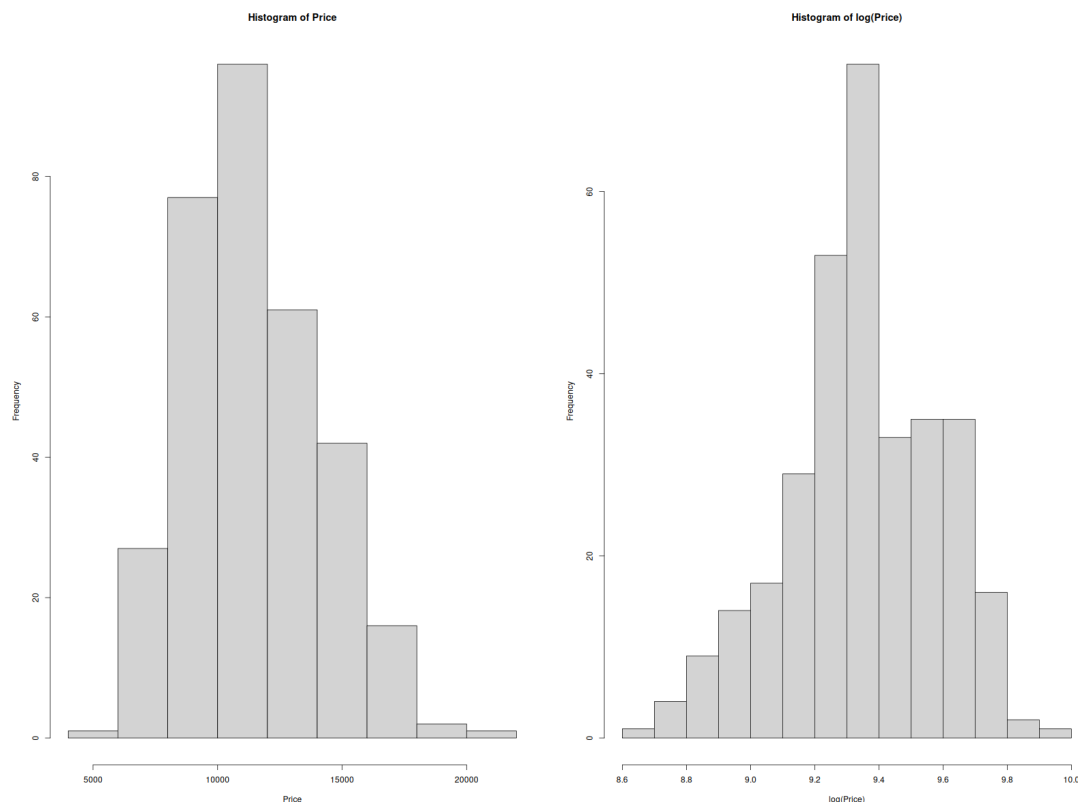


Figure 4: Histograms of Normal Price (left) and Log-Transformed Price (right)

The normal histogram (Figure 4, left) appears right-skewed, while the log-transformed histogram (Figure 4, right) appears more symmetric although somewhat left-skewed.

2.g Five Number Summary Without Outliers

	Original Price	Log-Transformed Price
Minimum	5995	5995.0
First Quartile (Q1)	9967.5	9950.0
Median (Q2)	11000	10999.5
Third Quartile (Q3)	14500	13800.0
Maximum	20995	18995.0

We see that the minimum remains the same, as no outliers were removed on the lower end. Conversely, our maximum decreased significantly after removing the upper outlier, indicating that the highest price was indeed an outlier. We see our first and third quartiles, along with the median, also decrease, reflecting the overall shift in the distribution after outlier removal.

3 Code Appendix

Listing 1: R Code for Lab 1

```

1 data <- read.csv("~/notes/math/2480/labs/lab1/data.csv")
2
3
4 summary(data)
5 # 1: Single categorical variable
6 # a) Frequency and relative frequency table for Color
7 color_freq <- table(data$Color)
8 color_rel_freq <- prop.table(color_freq)
9 color_table <- data.frame(
10     Color = names(color_freq),
11     Frequency = as.vector(color_freq),
12     RelativeFrequency = as.vector(color_rel_freq)
13 )
14 color_table
15
16 # b) Bar chart of Color frequencies
17 barplot(
18     color_freq,
19     main = "Car_Color_Frequency",
20     xlab = "Color",
21     ylab = "Frequency",
22     col = "steelblue"
23 )
24
25 # 2: One quantitative variable
26
27 # a) Mean and variance of Price
28 price_mean <- mean(data$Price, na.rm = TRUE)
29 price_var <- var(data$Price, na.rm = TRUE)
30 price_mean
31 price_var
32
33 # b) Five-number summary for Price
34 price_fivenum <- fivenum(data$Price, na.rm = TRUE)
35 price_fivenum
36
37 # c) Box plot of Price and outliers
38 price_box <- boxplot(data$Price, main = "Boxplot_of_Price", ylab
39     = "Price")
40 price_outliers <- price_box$out
41 price_outliers
42
43 # d) Log transform of Price (new column named "log(Price)")
44 data$'log(Price)' <- log(data$Price)
45
46 # e) Box plot of log(Price) and outliers
47 log_price_box <- boxplot(data$'log(Price)', main = "Boxplot_of_
48     log(Price)", ylab = "log(Price)")

```

```
47 log_price_outliers <- log_price_box$out
48 log_price_outliers
49
50 # f) Histograms for Price and log(Price)
51 hist(data$Price, main = "Histogram of Price", xlab = "Price", col
    = "lightgray")
52 hist(data$log(Price), main = "Histogram of log(Price)", xlab =
    "log(Price)", col = "lightgray")
53
54 # g) Five-number summary for Price with outliers removed
55 price_no_outliers <- data$Price[!(data$Price %in% price_outliers)
    ]
56 price_no_outliers_fivenum <- fivenum(price_no_outliers, na.rm =
    TRUE)
57 price_no_outliers_fivenum
```